(c) Broken or cracked main or side rods must not be welded by the Autogenous Process.

(d) Main and side rods must be maintained in accordance with Locomotive Maintenance Instructions Nos. L-17 and L-47.

(e) Welding and straightening of main and side rods must be in accordance with Locomotive Maintenance Instructions No. L-26.

(f) The side play of main and side rods on any one pin must not exceed ¼".

(g) Oil and grease cups must be properly maintained. See Locomotive Maintenance Instructions No. L-28.

(h) The pressures in tons per inch length of rod fit for applying pin bearing bushings to main and side rods are given on standard tracing F-66814, and also in Locomotive Maintenance Instructions No. L-47, for Side Rods.

25. **Classification Lamps:** Classification and marker lamps must be provided as required by Circular 95. These lamps must be kept clean.

26. **Smoke Boxes, Ash Pans and Ash Pan Rigging:** For instructions covering Inspection, Maintenance and Records see Locomotive Maintenance Instructions No. L-48.

27. **Axles:** (a) Driving, trailing and engine truck axles with any of the following defects must not be continued in service: Bent axle, cut journals that cannot be made to run cool without turning; seamy journals; unsafe on account of usage, accident or derailment; axles under the limits shown in Circular No. 105.

(b) Tender truck axles and trailing truck axles with outside journals must not be continued in service with the following defects: Bent axle, cut journals that cannot be made to run cool without turning; seamy journals; unsafe on account of usage, accident or derailment; collars broken or worn to ¼" or less in thickness; fillet in back shoulder worn out; axles under the limits shown in Circular No. 105.

(c) The place and date applied, the original diameter of journal and the kind of material must be stamped on the end of all new axles applied to locomotives and tenders in accordance with tracing No. D-412078. This marking covering the first application must not be obliterated during the life of the axle.

28. **Crank Pins:** (a) Crank pins must be securely applied. Shimming or prick punching crank pins must not be allowed. All new crank pins applied must have the place and date applied, original diameter and kind of material used, stamped on the end of the pin in accordance with tracing No. D-412078.

(b) An internal inspection of all hollow main crank pins must be made for detail fractures at the monthly and annual inspection periods. When the side rods are removed, an external inspection of all crank pins must be made for detail fractures, giving special attention to the back fillet next to the wheel hub. The minimum radius of the back fillet next to the wheel hub shall be ⅛" at class repairs.

(c) Crank pins must be renewed when their rod journals are worn to the following limits:

	Road Limit	Shop Limit
Diameter of rod journal, less than standard	¼"	7/32"
Rod journal, out of round	1/16"	1/32"
Rod journal, tapers	1/16"	1/32"
Rod journal, longer than standard	3/16"	⅛"
Main rod journal, eccentric	⅛"	3/32"

(d) For crank pin mounting pressure, see Locomotive Maintenance Instructions No. L-15.

29. **Driving Boxes and Cellars:** Driving boxes must be maintained in accordance with Locomotive Maintenance Instructions No. L-16. Broken crown bearings must be renewed. Loose crown bearings must be renewed or tightened with not more than one shim. Cellars must be maintained tight in the box. Worn cellar bolt holes must be repaired as shown on tracing No. F-96504.

30. **Driving Box Shoes and Wedges:** When necessary to apply liners to driving box shoes or wedges, a single liner of sufficient thickness, instead of two or more thin liners, must be riveted to each shoe or wedge. The minimum thickness of the wall to which the liner is riveted must be ½" after repairs are completed. Phosphorous bronze shoes that are in good condition except for short cracks at the corners or on the flange may be reclaimed by welding as per tracing No. F-90228.